Class 13: Evidence from CAPM and APT V

March 3, 2026

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Today's Question

If markets are efficient, why do behavioral biases persist?

What limits arbitrageurs from correcting mispricings?

Positioning

Where We've Been	Where We Are	Where We're Going
Risks vs. opportunities, the factor zoo, and market efficiency (all three forms)	The behavioral critique and limits to arbitrage	Active portfolio management and the Treynor-Black model

Learning Objectives

By the end of today's class, you should be able to:

1. Describe the key behavioral biases: overconfidence, representativeness, and inattention
2. Explain the concept of "limits to arbitrage" and why mispricings can persist
3. Articulate the tension between the EMH and behavioral finance

The Behavioral Critique

In contrast to the EMH, the behavioral view suggests prices may be wrong.

Two departures from the EMH:

1. Irrational investors
2. Limits to arbitrage

This leads to investors who make systematic errors in pricing securities, and ensures that attempts to exploit/correct mispricings are risky or costly.

The Behavioral Critique

Let's begin with "investor irrationality."

Some key ideas about how investors may behave irrationally:

- Overconfidence
- Representativeness
- Inattention

Overconfidence

Two types:

1. **Overprecision**
   - People are too confident in the accuracy of their beliefs
   - 90% confidence intervals contain correct answers less than 50% of the time

2. **Overplacement**
   - Overly rosy views of their abilities
   - In surveys, 90% believe themselves to be above the median on various dimensions

Overplacement: Quiz

Overconfidence: Applications

View in finance research: excessive trading by individual (retail) investors.

Barber and Odean (2002) show male traders trade more and do worse:

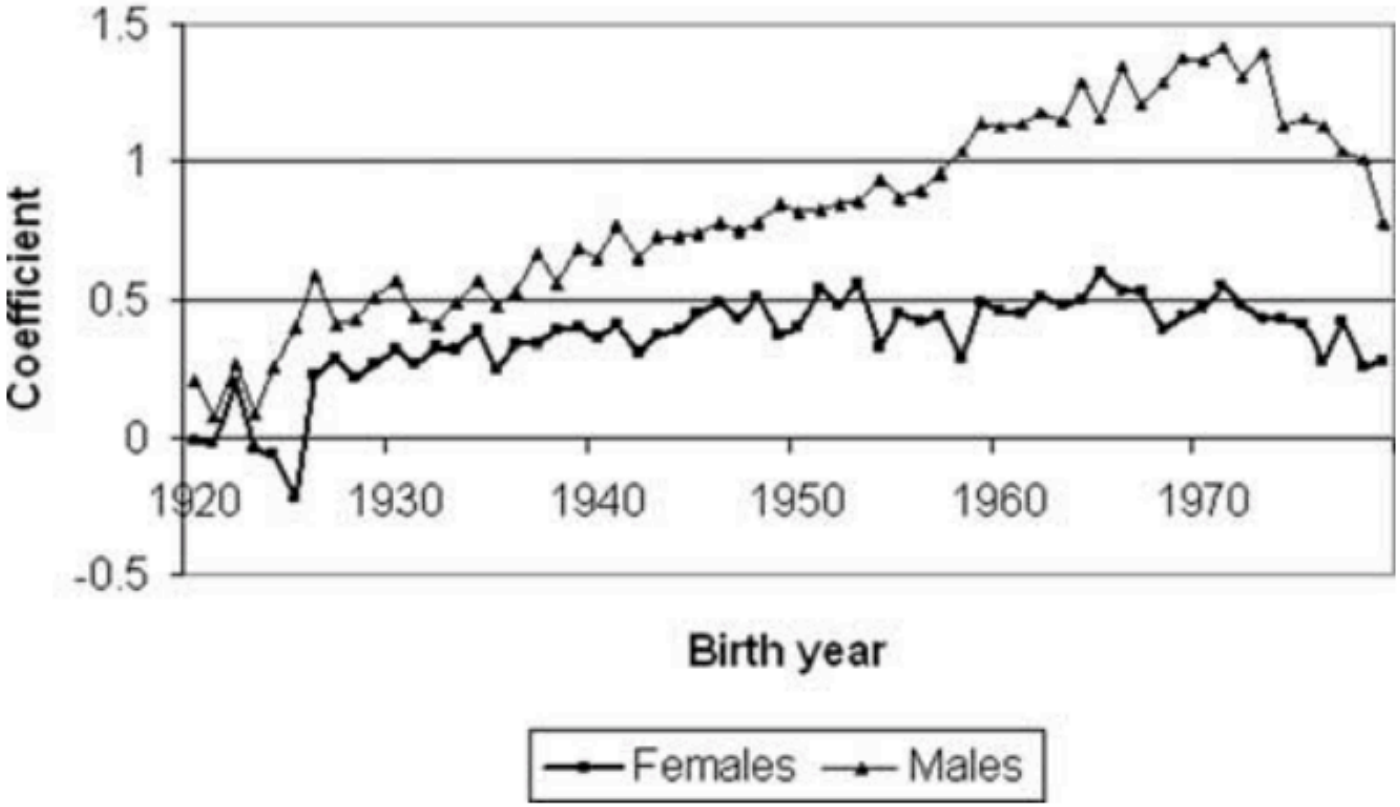
- Psychological studies suggest men more prone to overconfidence

Grinblatt and Keloharju (2009) combine trading data with psychological tests of overconfidence:

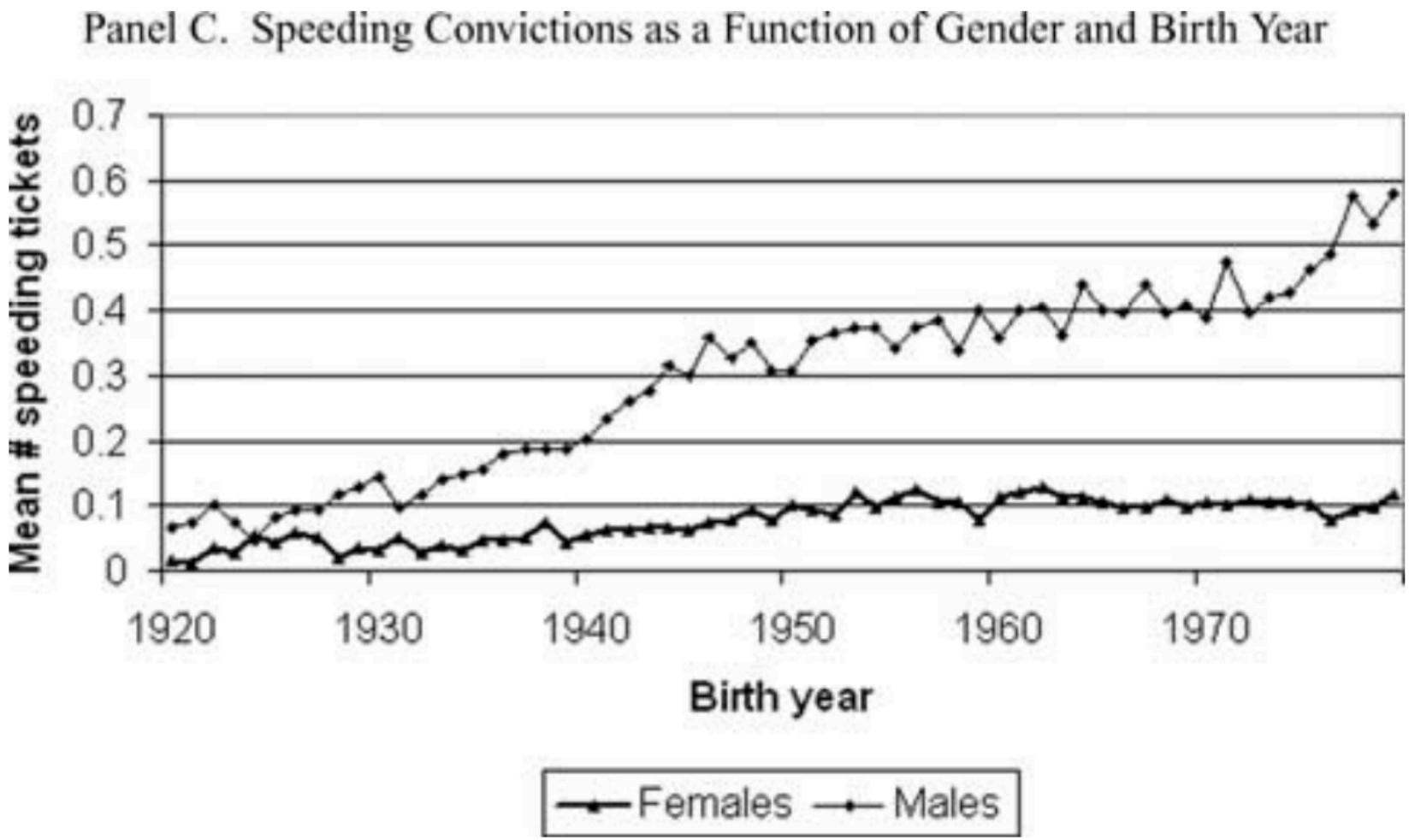
- Overconfidence linked to intensity of trading activity

Overconfidence: Gender and Trading

Panel B. Marginal Effects of Gender and Birth Year on Average Number of Trades with Effects of Control Variables Taken Out



Overconfidence: Gender and Speeding



Representativeness

People extrapolate past trends too far into the future.

Just because a firm has seen high growth for the past five years does not necessarily mean it will continue.

In revising their beliefs, individuals tend to overweight recent information and underweight prior (or base rate) data:

- *"The predicted value is selected so that the standing of the case in the distribution of outcomes matches its standing in the distribution of impressions"* (Kahneman and Tversky)

Representativeness: Applications

Representativeness predicts long-run mean reversion in stock returns (De Bondt and Thaler, 1985):

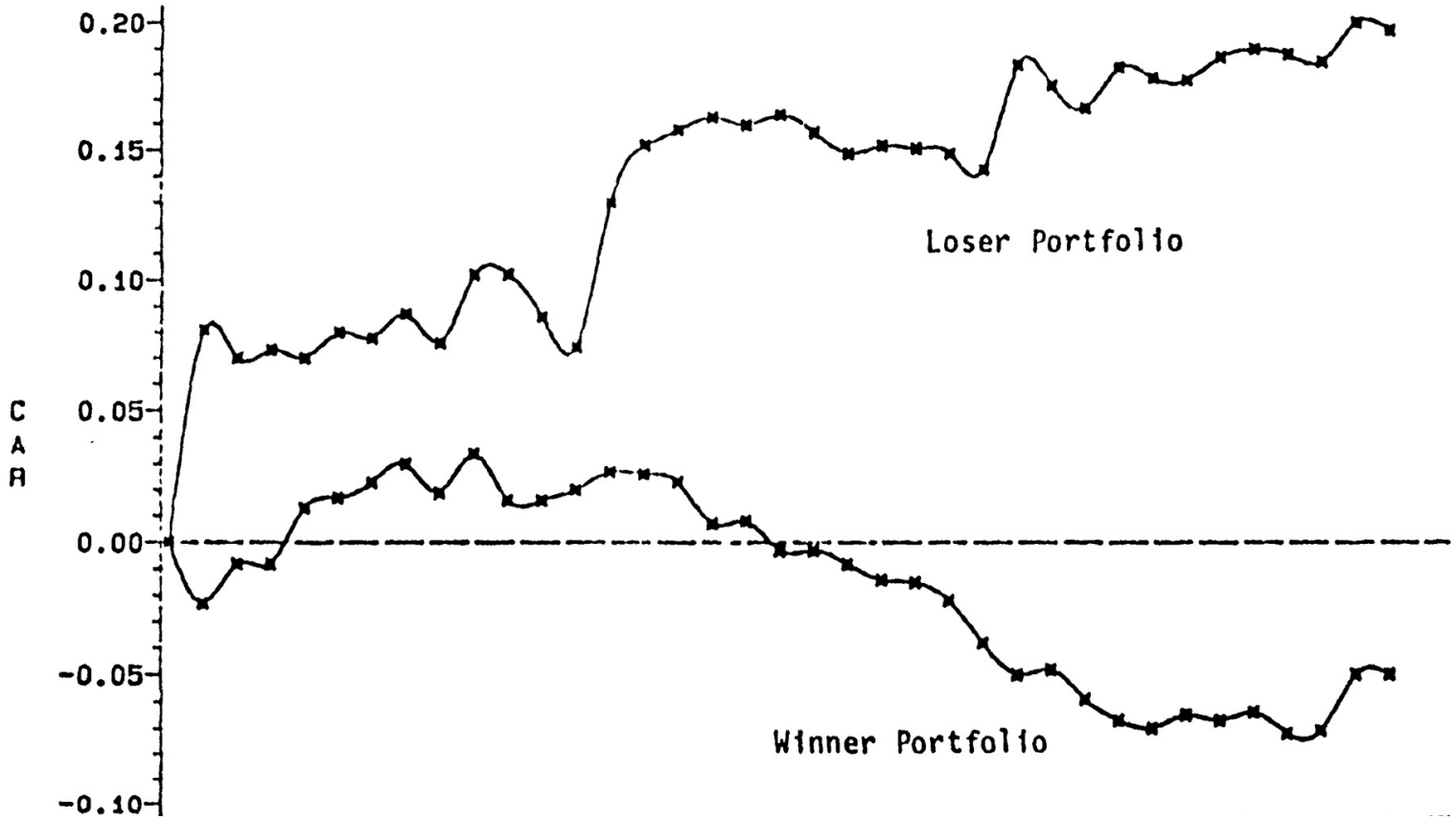
Exercise:

1. Take a three-year period and rank stocks on their performance
2. Form a "losers" ("winners") portfolio of the 30 worst-performing (best-performing) stocks, and hold for three years
3. Repeat process three years later

Result: past winners do poorly, past losers rebound.

- Note: observe short-term momentum, but long-run reversals

Long-Run Reversals (De Bondt and Thaler 1985)



Other Examples of Over-Extrapolation

Value premium may reflect over-extrapolation about future growth rates.

Low book-to-market firms priced based on expectations that recent growth will continue.

Growth stocks tend to disappoint in earnings announcements.

Earnings disappointments more severe for growth stocks.

Earnings Surprises

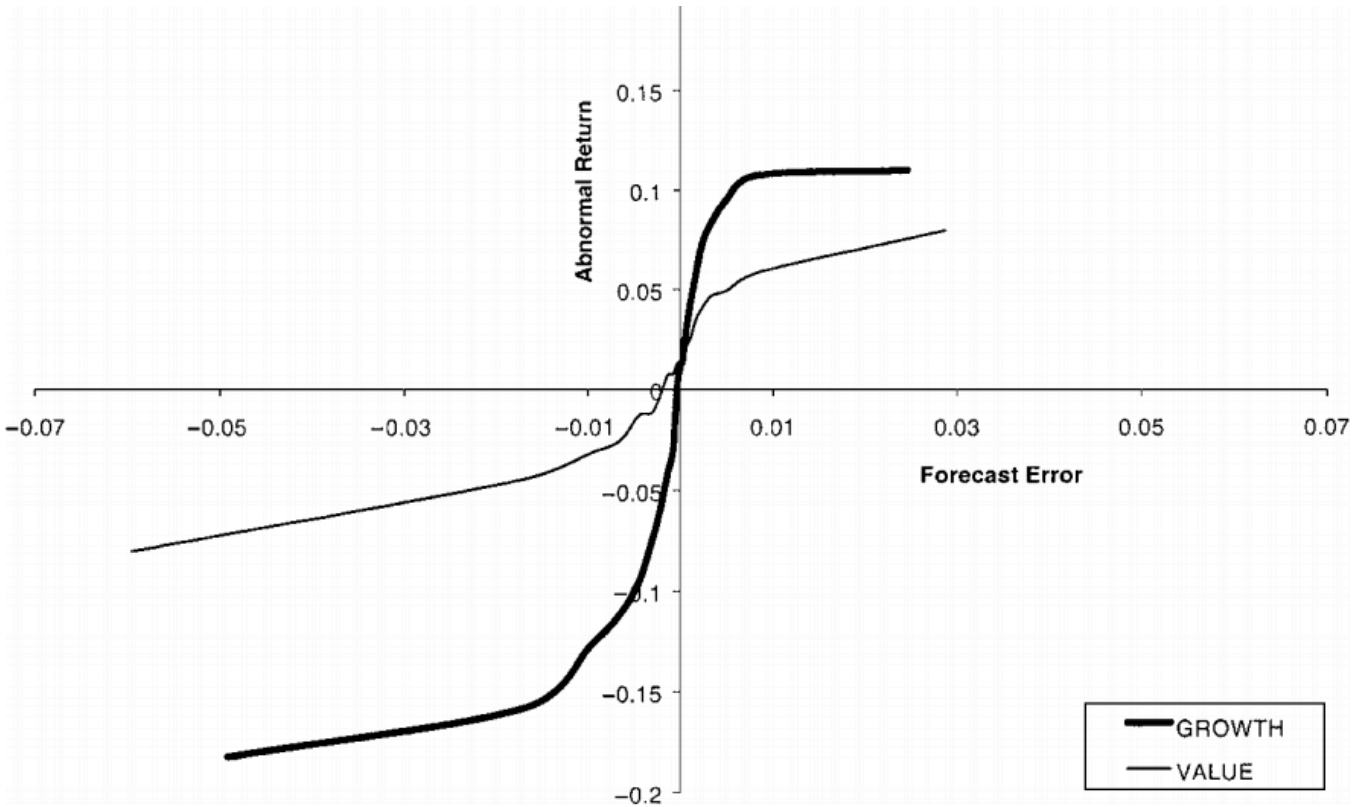


Figure 4. Earnings surprise response functions for value and growth stocks. This graph plots the quarterly abnormal returns for value and growth stocks respectively as a function of the magnitude of the quarterly earnings forecast error. Each plot is formed by dividing the stocks into 20 portfolios based on the magnitude of the forecast error, and then plotting the mean portfolio abnormal returns and forecast errors. The resulting points are joined using

Inattention

People have limited capacity for gathering and processing information.

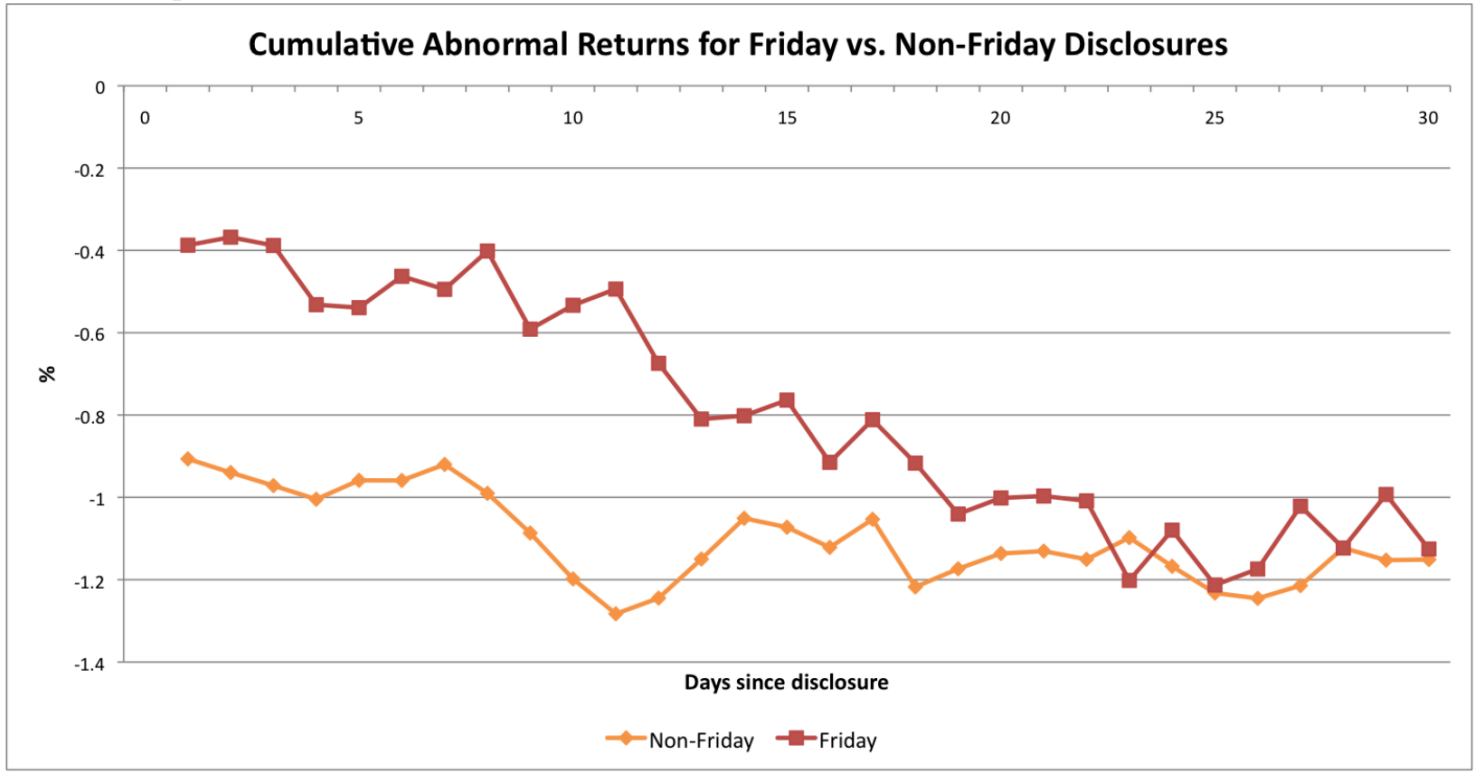
Application: Post-earnings announcement drift

- After earnings surprises, returns drift for as long as 60 days
- Larger for Friday announcements
- Larger for firms announcing at same time as other firms

See also Niessner (2014), Cohen and Frazzini (2008).

Niessner (2014)

Small Companies:



Cohen and Frazzini (2008)

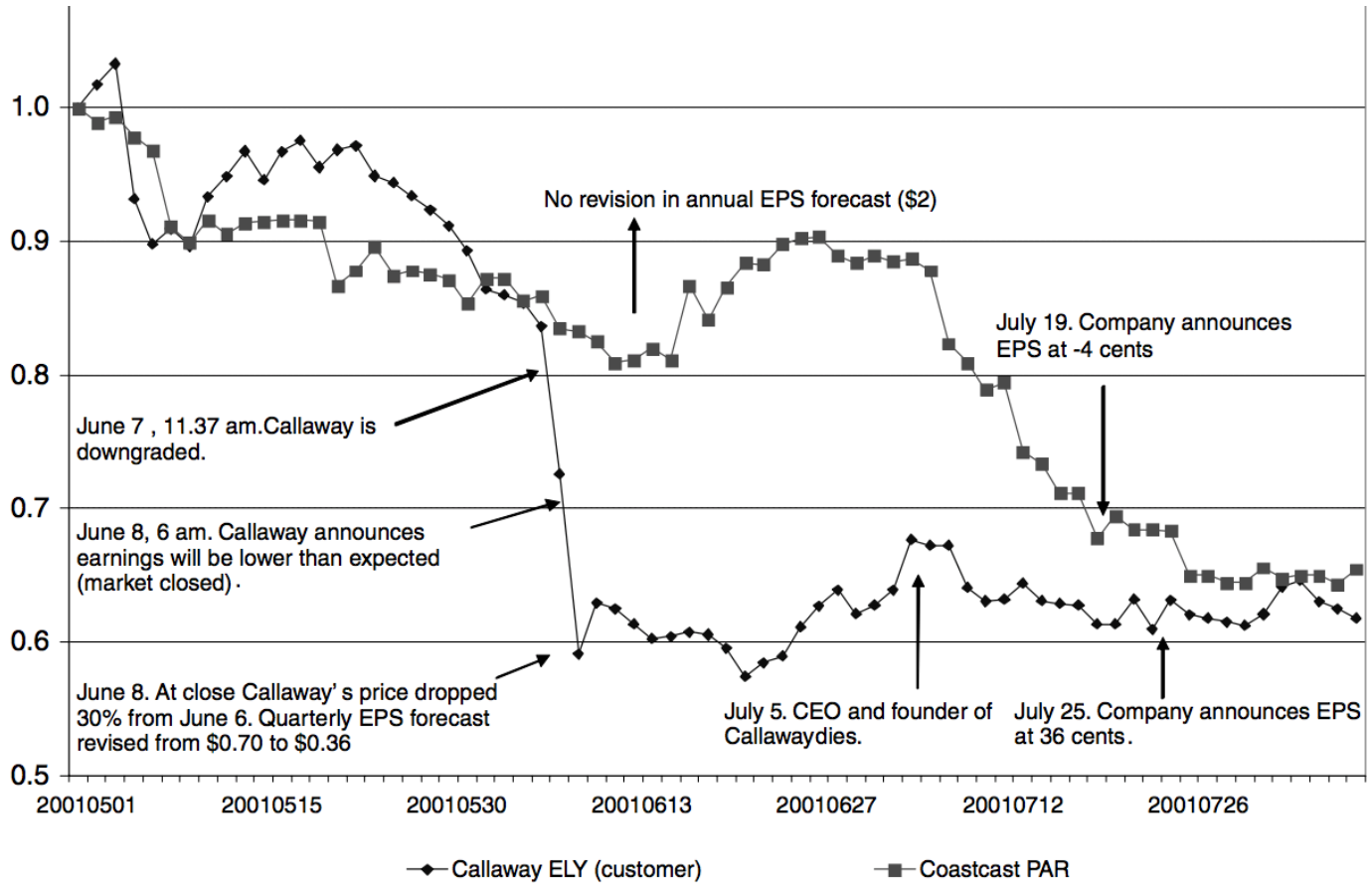


Figure 1. Coastcast Corporation and Callaway Golf Corporation. This figure plots the stock prices of Coastcast Corporation (ticker = PAR) and Callaway Golf Corporation (ticker = ELY)

Why Don't Arbitrageurs Exploit Mispricing?

The second piece of the behavioral framework — “limits to arbitrage.”

Recall our willingness to take positions is driven by not just return, but by the risk associated with the reward.

If an investment has α, it may also include $\sigma(\varepsilon)$

Why Don't Arbitrageurs Exploit Mispricing?

Royal Dutch and Shell Transport merge interests, but continue to trade separately in the Netherlands and the UK.

Note: a long-short position is risky (why?)

- Behavioral biases generating mispricing may get worse, not better

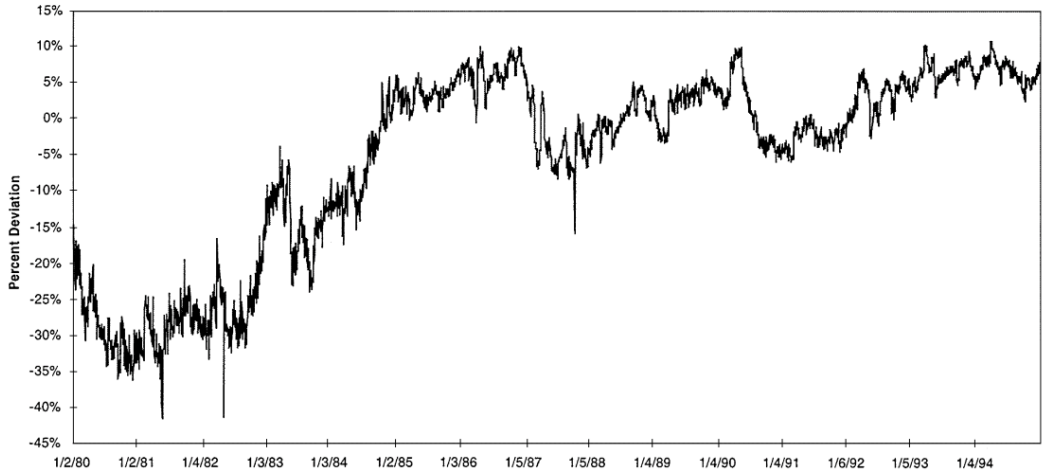


Fig. 1. Log deviations from Royal Dutch/Shell parity. Note: This figure shows on a percentage basis the deviations from theoretical parity of Royal Dutch and Shell shares and ADRs traded on the NYSE. Data are from the Center for Research in Security Pricing (CRSP).

Behavioral Finance and the EMH

If markets are efficient, why do behavioral biases persist?

What limits arbitrageurs from correcting mispricings?

Anomalies reflect a failure in the efficient markets / rational investor model.

Behavioral finance may provide answers, based on known biases from cognitive psychology.

Next Class: March 24

Topics: Active Portfolio Management

- Portfolio construction
- The Treynor-Black model
- Information ratios

Note: Spring Break March 7–22